



**SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)**

Constituent of Symbiosis International (Deemed University), Pune

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## **ASSIGNMENT No: 8**

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**BATCH: CSE B3**

**A.Y: 2025-29**

### **TITLE:**

Write a Java program to demonstrate the use of the super keyword in inheritance.

### **Problem Statement**

Write a Java program to demonstrate the use of the super keyword in inheritance.

### **Learning Objectives**

To demonstrate the use of the super keyword for accessing parent class members.

### **Theory**

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods.

Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding

### **Exercise**

1. Create an **Employee and Manager program** where Manager accesses Employee details using super keyword.

### **SOURCE CODE:**

```

class Employee {
    String name;
    int empId;

    Employee(String name, int empId) {
        this.name = name;
        this.empId = empId;
    }

    void display() {
        System.out.println("Employee Name: " + name);
        System.out.println("Employee ID: " + empId);
    }
}

class Manager extends Employee {
    String department;

    Manager(String name, int empId, String department) {
        super(name, empId); // Calls Employee constructor
        this.department = department;
    }

    void showDetails() {
        super.display(); // Calls Employee display method
        System.out.println("Department: " + department);
    }
}

Run | Debug
public static void main(String[] args) {
    Manager m = new Manager(name: "Suhani", empId: 101, department: "HR");
    m.showDetails();
}
}

```

## OUTPUT:

```

suhanigarg670@Suhani-MacBook-Air ASSIGNMENT 8 % cd "/Users/suhanigarg670/JAVA LAB/ASSIGNMENT 8/" && javac Manager.java && java Manager
Employee Name: Suhani
Employee ID: 101
Department: HR
suhanigarg670@Suhani-MacBook-Air ASSIGNMENT 8 %

```

2. Develop a **Vehicle Insurance System** where child insurance classes access parent vehicle information using the super keyword.

### SOURCE CODE:

```
class Vehicle {
    String vehicleNo;
    String ownerName;

    Vehicle(String vehicleNo, String ownerName) {
        this.vehicleNo = vehicleNo;
        this.ownerName = ownerName;
    }

    void displayVehicle() {
        System.out.println("Vehicle Number: " + vehicleNo);
        System.out.println("Owner Name: " + ownerName);
    }
}

class Insurance extends Vehicle {
    String policyNo;

    Insurance(String vehicleNo, String ownerName, String policyNo) {
        super(vehicleNo, ownerName); // Calls Vehicle constructor
        this.policyNo = policyNo;
    }

    void displayInsurance() {
        super.displayVehicle(); // Calls parent method
        System.out.println("Policy Number: " + policyNo);
    }
}

Run | Debug
public static void main(String[] args) {
    Insurance obj = new Insurance(vehicleNo: "MH12AB1234", ownerName: "Suhani", policyNo: "P101");
    obj.displayInsurance();
}
```

### OUTPUT:

```
● suhanigarg670@Suhani-MacBook-Air ASSIGNMENT 8 % cd "/Users/suhanigarg670/JAVA LAB/ASSIGNMENT 8/" && javac Insurance.java && java Insurance
Vehicle Number: MH12AB1234
Owner Name: Suhani
Policy Number: P101
○ suhanigarg670@Suhani-MacBook-Air ASSIGNMENT 8 %
```

**Conclusion:**

The program demonstrated the use of the super keyword in Java inheritance. It was used to call the parent class constructor and access parent class methods and variables from the child class. This helps establish proper constructor chaining, promotes code reuse, and ensures effective implementation of inheritance in Java.